Main Interface. The primary view of ProtoMech displays a grid layout where rows correspond to CLT layers and columns correspond to sequence positions (Figure 9). Each cell contains the latents that activate at that position, with activation intensity encoded by color. When virtual weights are enabled, edges connect latents across layers, with blue indicating positive weights and orange indicating negative weights. The protein sequence is displayed along the bottom axis, allowing users to directly relate latent activations to specific amino acid residues.

Exploring Latent Rankings. Clicking on a layer label (e.g., "Layer 2") opens the Latent Rankings panel, which displays the top-activating latents for that layer ranked by their maximum activation value (Figure 10). Each entry shows the latent index, maximum activation score, the sequence position where maximum activation occurs, and a sequence context window with the activation site highlighted. Users can directly add latents to the canvas or access detailed feature information from this panel.

Latent Information Panel. Clicking on any latent opens a detailed information panel with three tabs (Figures 11–13):

- *Sequences* (Figure 11): Displays the input sequence with a heatmap overlay showing activation intensities across all positions. The current position's amino acid and activation value are shown at the top. Below, the top-activating sequences from Swiss-Prot are listed with their activation scores, allowing users to identify common patterns that drive latent activation.
- *Alignment* (Figure 12): Presents an alignment view where the input sequence is aligned with the top-activating sequences from Swiss-Prot. Activation intensities are overlaid as colored highlights, enabling users to identify conserved motifs and structural features that the latent has learned to recognize.
- *Influences* (Figure 13): Lists all incoming connections (from earlier layers) and outgoing connections (to later layers) for the selected latent. Each connection displays the source/target layer and latent index, the virtual weight magnitude (positive in green, negative in orange), and the number of sequence positions where this connection is active. This view enables systematic exploration of how information flows through the circuit.

Building and Analyzing Circuits on the Canvas. The canvas provides an interactive workspace for constructing and visualizing circuit diagrams (Figure 14). Users can add latents to the canvas by right-clicking on any latent in the grid view or using the "Add to Canvas" button in the Latent Info panel. Once on the canvas, nodes can be freely repositioned and annotated with descriptive labels (e.g., "Glycine-rich loop", "ATP-binding site") to document hypothesized functions. Virtual weight edges are automatically drawn between canvas nodes, with edge color indicating sign (blue for positive, orange for negative) and thickness proportional to magnitude. The "Filter Virtual Weights" slider controls the minimum weight threshold for displayed edges. Canvas layouts can be saved and loaded for reproducibility, and exported as images for publication.

Interpreting Virtual Weights. The edge weight shown between two latents on the canvas represents the average virtual weight across all sequence positions where both latents are active. For example, if latents L1/1983 and L5/1774 co-activate at multiple positions with varying weights, the displayed edge weight is their mean. Users can click the "View" button in the Influences tab to examine position-specific weights in detail.

G. Comparison to Sparse Autoencoders

SAEs differ from ProtoMech in that they are optimized for reconstruction, rather than transcoding computation. Fig. 15 shows a simplistic example of the differences between SAEs and transcoders. An SAE operating at layer ℓ is trained to take as an input $\mathbf{y}^\ell$ and reconstruct $\hat{\mathbf{y}}^\ell$. In contrast, ProtoMech is designed to take as an input $\mathbf{x}^\ell$ and compute $\hat{\mathbf{y}}^\ell$. This means that SAEs, by design, are ignoring the MLP computations inside the transformer, making SAEs insufficient to be replacement model.

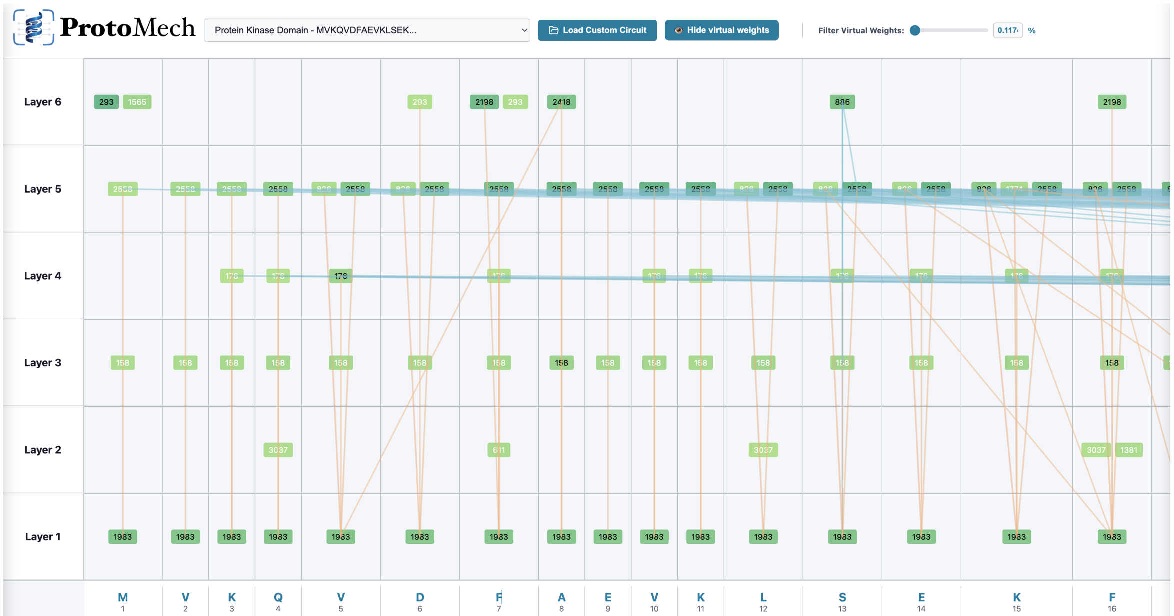

Figure 9. **ProtoMech main interface.** The grid view displays latent activations organized by layer (rows) and sequence position (columns). Each colored box represents an active latent, with the latent index shown inside. When virtual weights are enabled, edges connect latents across layers: blue edges indicate positive weights (activation) and orange edges indicate negative weights (inhibition). The protein sequence is displayed along the bottom axis. The toolbar provides access to example circuits, custom data loading, and virtual weight filtering controls.

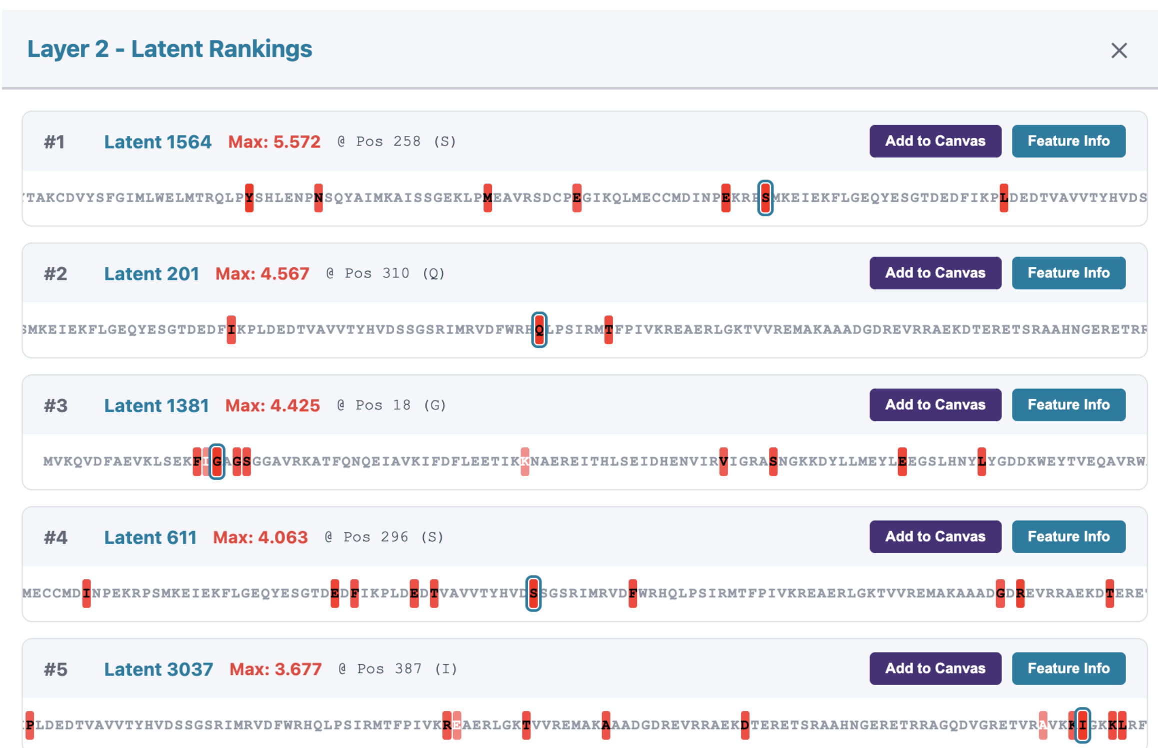

Figure 10. **Layer-wise latent rankings.** Clicking on a layer label opens this panel, showing the top-activating latents for that layer ranked by maximum activation value. Each entry displays the latent index, maximum activation score, the sequence position of peak activation, and a sequence window centered on the activation site (highlighted in red). The “Add to Canvas” button enables quick circuit construction, while “Feature Info” opens the detailed latent information panel.

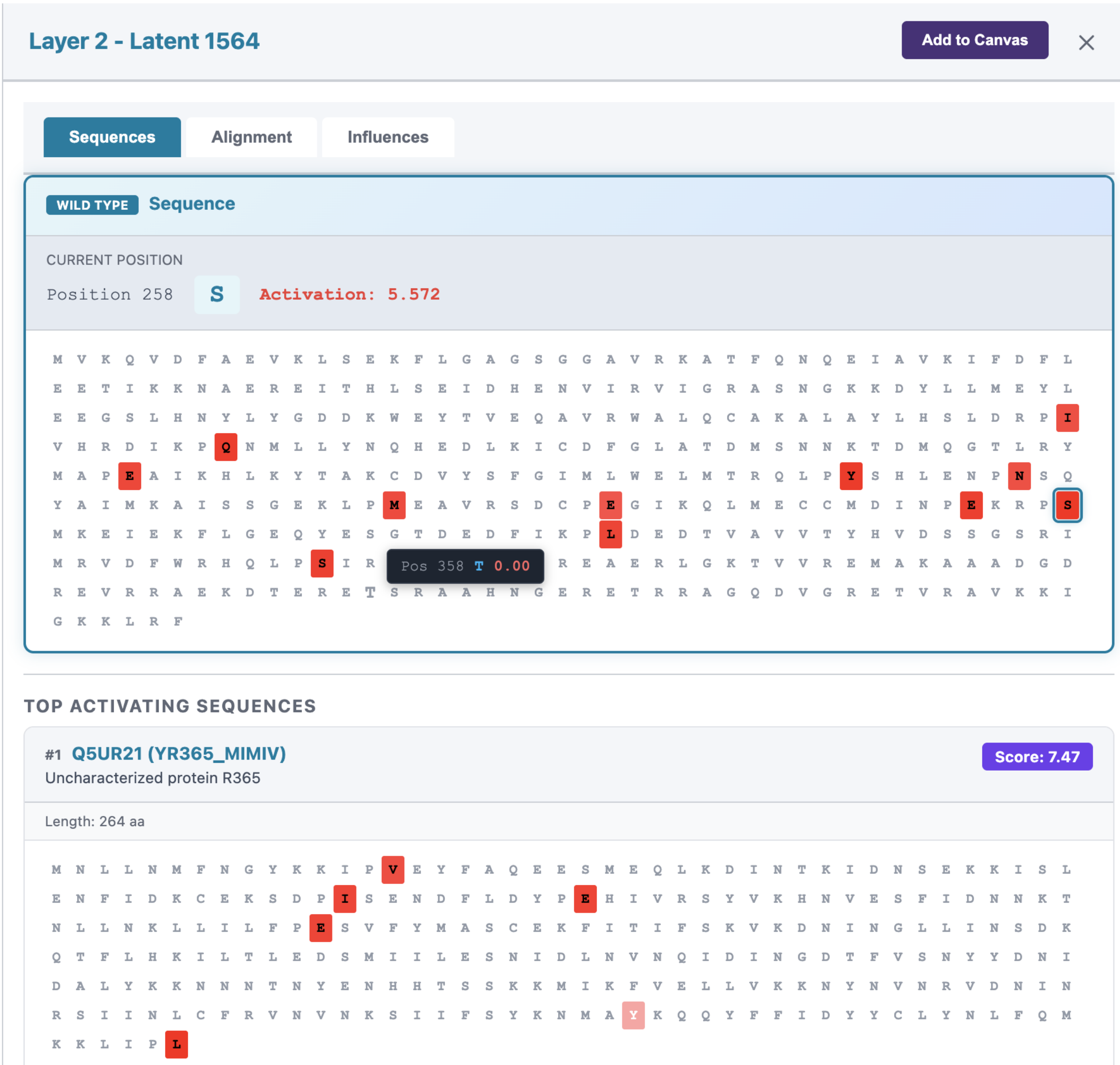

Figure 11. **Latent Information Panel: Sequences tab.** The Sequences tab provides detailed activation information for a selected latent. The top section shows the input sequence with activation intensities displayed as a heatmap, where darker colors indicate stronger activation. The current position, amino acid, and activation value are displayed above the sequence. The bottom section lists top-activating sequences from Swiss-Prot, each with its UniProt identifier, protein name, and activation score, enabling users to identify sequence patterns that maximally activate the latent.